NKPC HSA Bayesian State-Space Estimation: Output gap, HP filter

June 23, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is

$$N_t^{model} = \frac{100\log(N_t^{level}) - \overline{100\log(N^{level})}}{10}.$$

Thus N_t^{model}, $\hat{N}_t$, and $\bar{N}_t$ are centered and measured in ten-log-point units. If a run metadata file records another `n_transform`, interpret the corresponding coefficients using that metadata.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + \kappa x_t - \theta\hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t - (\theta_0 + \gamma\bar{N}_t)\hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1\hat{N}_{t-1} + \rho_2\hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ, κ_0, and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata. Because the default N transform is centered and divided by 10, the reported δ, θ, θ_0, and γ coefficients are already effects for a ten-log-point movement in the corresponding N component. κ_0 and θ_0 are intercepts at average $\bar{N}_t$.

4 Coefficient Summary

Table 1 reports posterior means and 95% credible intervals for key parameters across all four models (baseline prior, full sample). Cells show *mean* [*ci_2.5*, *ci_97.5*].

Table 1: Posterior means and 95% credible intervals by model

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.670 [0.566, 0.773]	0.664 [0.561, 0.762]	0.637 [0.519, 0.750]	0.624 [0.507, 0.739]
kappa	0.138 [0.047, 0.227]	NaN	0.144 [0.047, 0.239]	NaN
kappa_0	NaN	0.135 [0.042, 0.225]	NaN	0.139 [0.035, 0.239]
delta	NaN	-0.011 [-0.044, 0.024]	NaN	-0.011 [-0.047, 0.024]
theta	NaN	NaN	0.168 [-0.216, 0.592]	NaN
theta_0	NaN	NaN	NaN	0.220 [0.015, 0.518]
gamma	NaN	NaN	NaN	-0.003 [-0.043, 0.036]
rho_1	NaN	0.755 [0.403, 1.044]	0.878 [0.507, 1.116]	0.967 [0.701, 1.208]
rho_2	NaN	-0.018 [-0.300, 0.233]	0.005 [-0.202, 0.216]	-0.051 [-0.276, 0.177]
phi_1	0.896 [0.821, 0.972]	0.895 [0.822, 0.968]	0.897 [0.827, 0.967]	0.897 [0.825, 0.972]
lambda_ez	0.134 [-0.093, 0.361]	0.127 [-0.098, 0.351]	NaN	0.099 [-0.131, 0.326]
n	NaN	-0.020 [-0.083, 0.041]	-0.021 [-0.070, 0.025]	-0.008 [-0.061, 0.044]

5 Result Blocks

Results are organized by condition. Each section fixes a data specification, prior specification, and sample period, then reports the coefficient table, prior/posterior overlay by model, Savage-Dickey density ratios, time-varying κ_t path, and model-comparison output for that condition.

6 output_gap_hp: prior=baseline, period=full

Condition. data=`output_gap_hp`, prior=`baseline`, period=`full`, constraint=`unrestricted` (unrestricted).

6.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.670 [0.566, 0.773]	0.664 [0.561, 0.762]	0.637 [0.519, 0.750]	0.624 [0.507, 0.739]
kappa	0.138 [0.047, 0.227]	NaN	0.144 [0.047, 0.239]	NaN
kappa_0	NaN	0.135 [0.042, 0.225]	NaN	0.139 [0.035, 0.239]
delta	NaN	-0.011 [-0.044, 0.024]	NaN	-0.011 [-0.047, 0.024]
theta	NaN	NaN	0.168 [-0.216, 0.592]	NaN
theta_0	NaN	NaN	NaN	0.220 [0.015, 0.518]
gamma	NaN	NaN	NaN	-0.003 [-0.043, 0.036]
rho_1	NaN	0.755 [0.403, 1.044]	0.878 [0.507, 1.116]	0.967 [0.701, 1.208]
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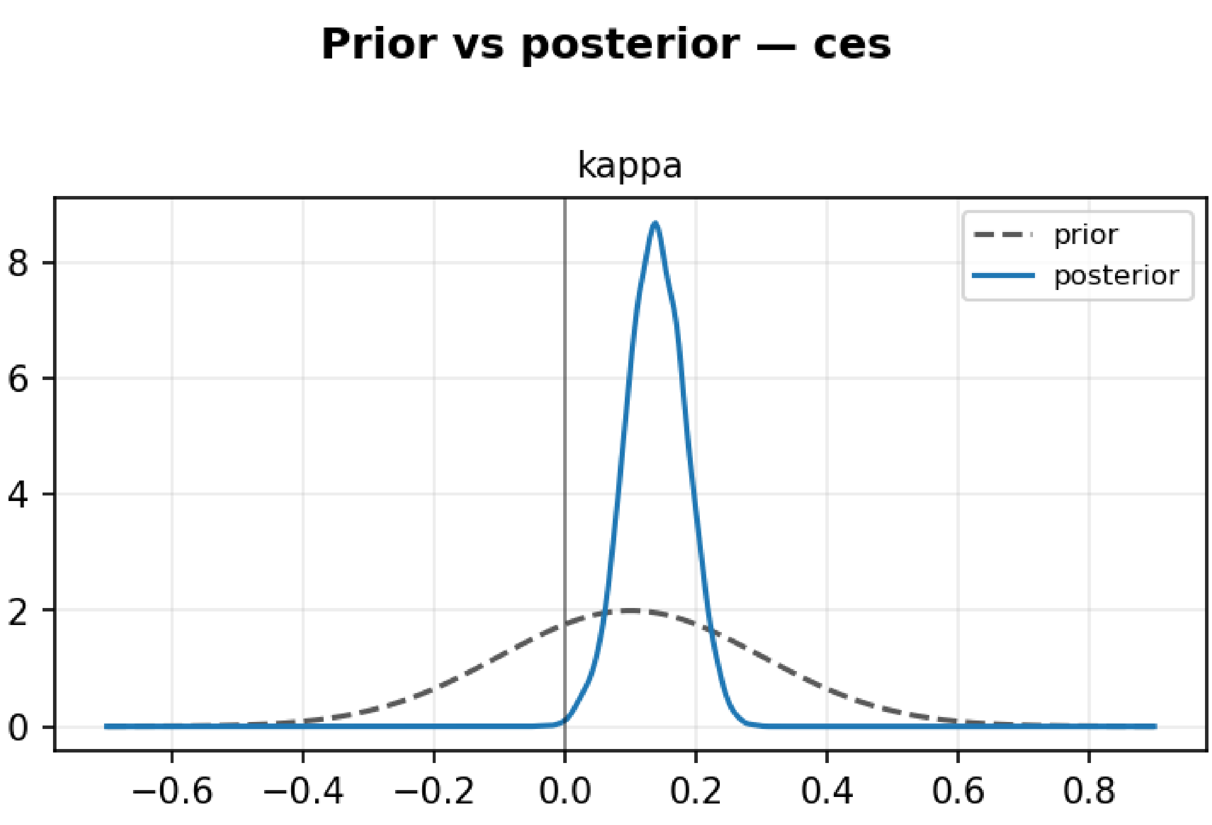
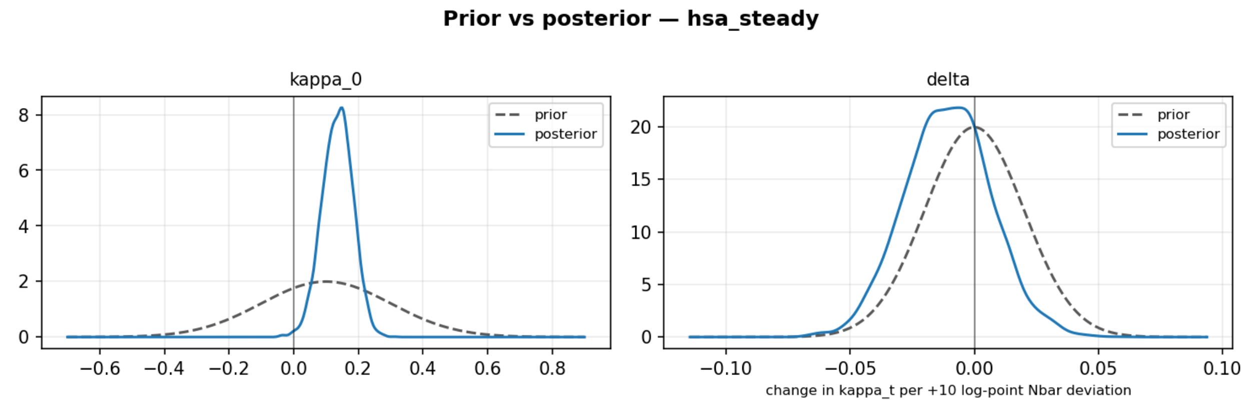
lambda_ez	0.134 [-0.093, 0.361]	0.127 [-0.098, 0.351]	NaN	0.099 [-0.131, 0.326]
n	NaN	-0.020 [-0.083, 0.041]	-0.021 [-0.070, 0.025]	-0.008 [-0.061, 0.044]

6.2 Prior vs Posterior by Model

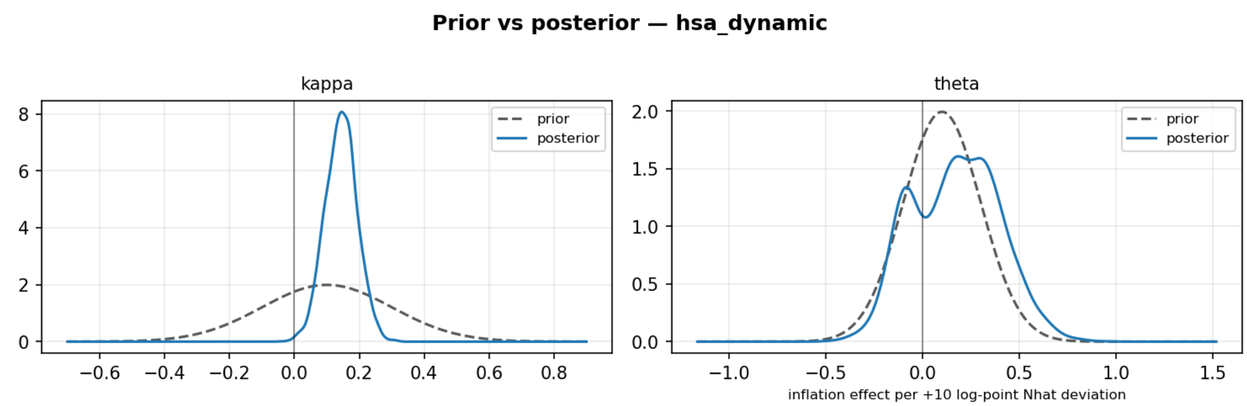
ces



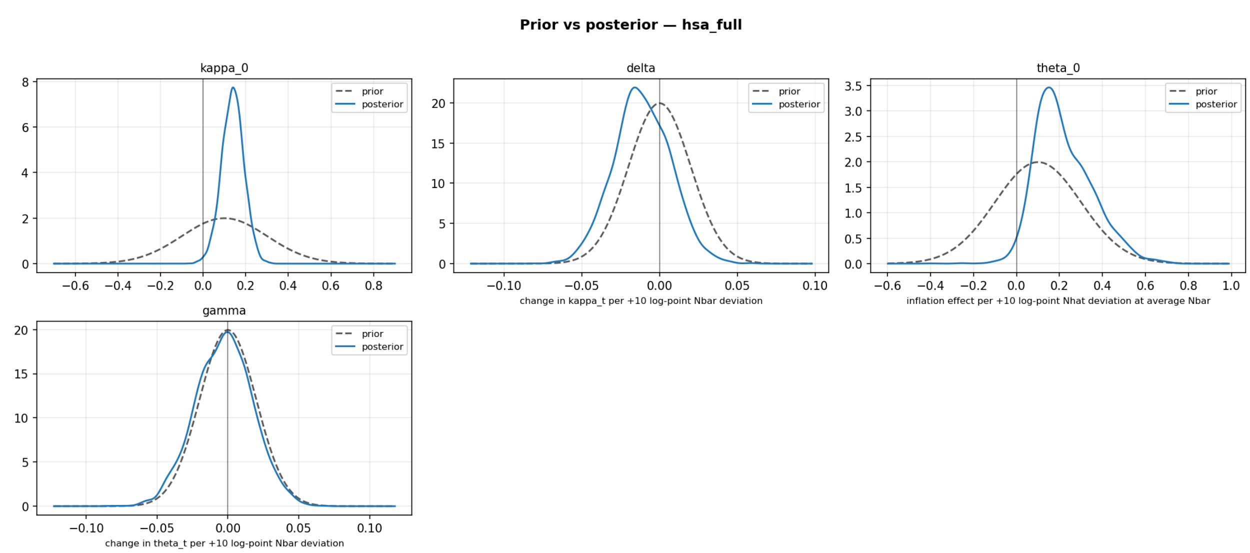
hsa_steady



hsa_dynamic



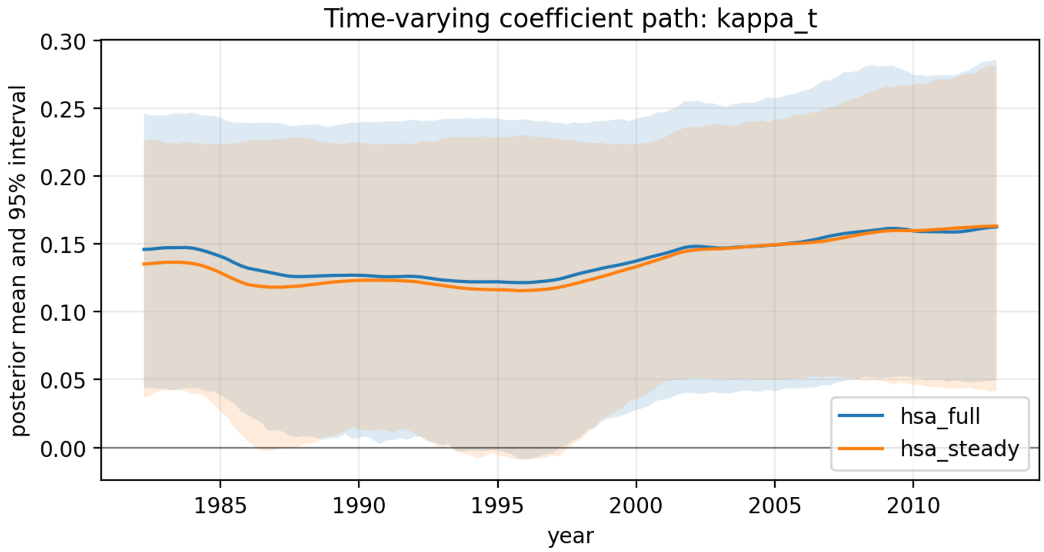
hsa_full

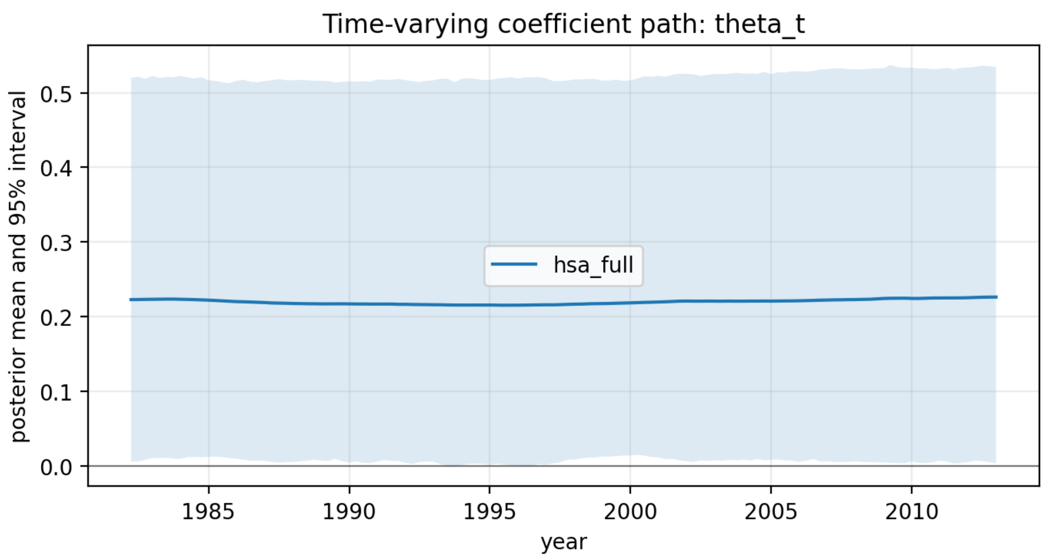


6.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0587
hsa_dynamic	kappa=0	0.1000	0.2000	0.1019
hsa_dynamic	theta=0	0.1000	0.2000	0.6200
hsa_full	kappa_0=0	0.1000	0.2000	0.1666
hsa_full	delta=0	0.0000	0.0200	0.8608
hsa_full	theta_0=0	0.1000	0.2000	0.2893
hsa_full	gamma=0	0.0000	0.0200	0.9883
hsa_steady	kappa_0=0	0.1000	0.2000	0.1246
hsa_steady	delta=0	0.0000	0.0200	1.0043

6.4 Time-Varying Kappa Path





6.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-114.0771	0.6059	-239.3121	1.0000	NaN	NaN	NaN
hsa_dynamic	-111.6291	0.5953	NaN	NaN	NaN	0.6200	NaN
hsa_full	-105.9814	0.5657	NaN	NaN	0.8608	NaN	0.9883
hsa_steady	-113.6351	0.6035	-302.1031	0.0000	1.0043	NaN	NaN

7 output_gap_hp: prior=baseline, period=full, constraint=kappa ¿= 0

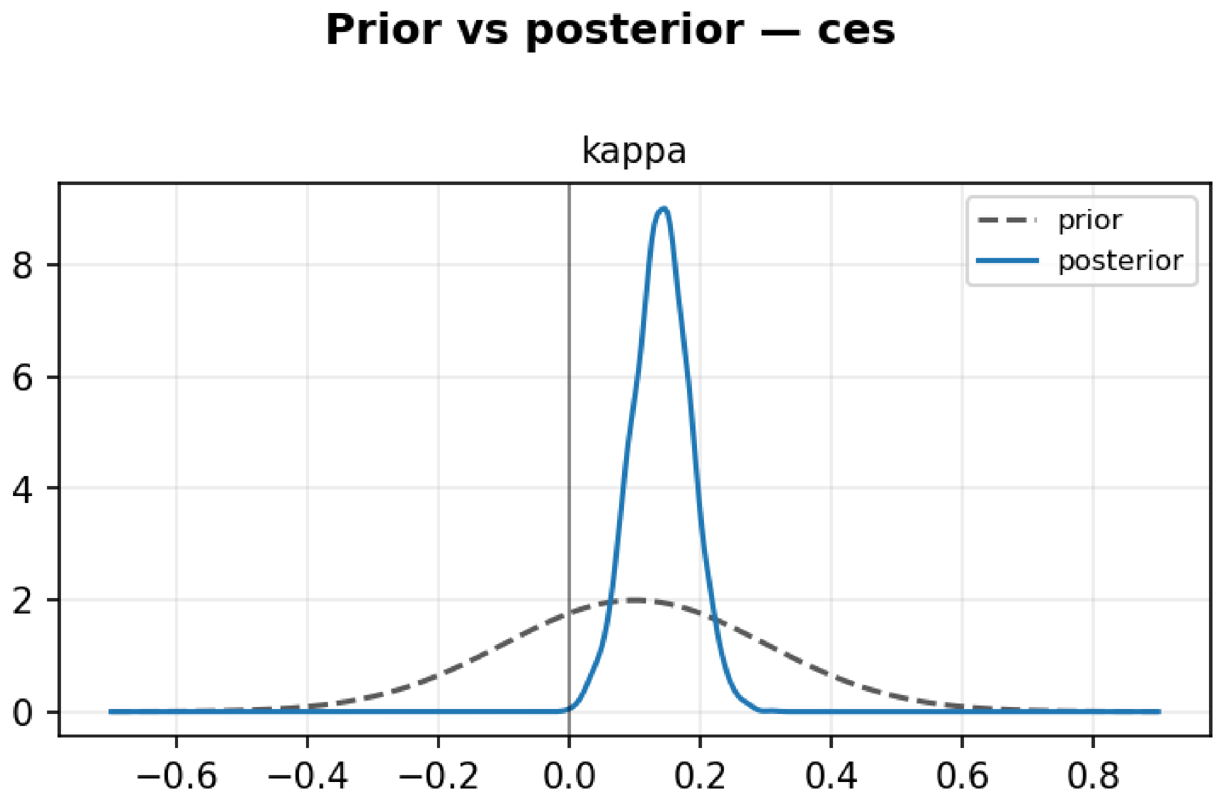
Condition. data=`output_gap_hp`, prior=`baseline`, period=`full`, constraint=`restricted_kappa` (kappa ¿= 0).

7.1 Coefficient Table

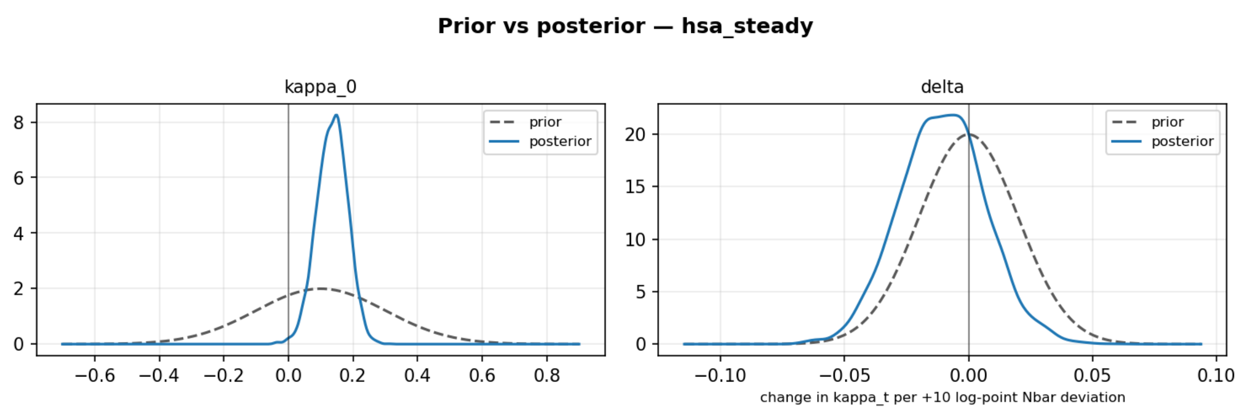
parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.672 [0.564, 0.776]	0.664 [0.561, 0.762]	0.639 [0.523, 0.747]	0.624 [0.507, 0.739]
kappa	0.140 [0.052, 0.227]	NaN	0.141 [0.050, 0.238]	NaN
kappa_0	NaN	0.135 [0.042, 0.225]	NaN	0.139 [0.035, 0.239]
delta	NaN	-0.011 [-0.044, 0.024]	NaN	-0.011 [-0.047, 0.024]
theta	NaN	NaN	0.149 [-0.223, 0.579]	NaN
theta_0	NaN	NaN	NaN	0.220 [0.015, 0.518]
gamma	NaN	NaN	NaN	-0.003 [-0.043, 0.036]
rho_1	NaN	0.755 [0.403, 1.044]	0.887 [0.524, 1.123]	0.967 [0.701, 1.208]
rho_2	NaN	-0.018 [-0.300, 0.233]	0.005 [-0.218, 0.216]	-0.051 [-0.276, 0.177]
phi_1	0.896 [0.821, 0.972]	0.895 [0.822, 0.968]	0.898 [0.831, 0.969]	0.897 [0.825, 0.972]
lambda_ez	0.131 [-0.093, 0.358]	0.127 [-0.098, 0.351]	NaN	0.099 [-0.131, 0.326]
n	NaN	-0.020 [-0.083, 0.041]	-0.023 [-0.072, 0.025]	-0.008 [-0.061, 0.044]

7.2 Prior vs Posterior by Model

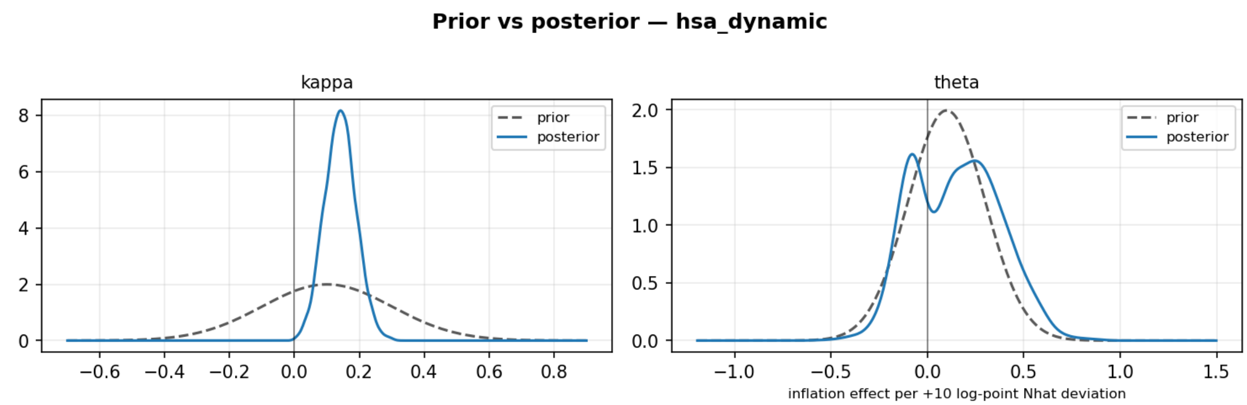
ces



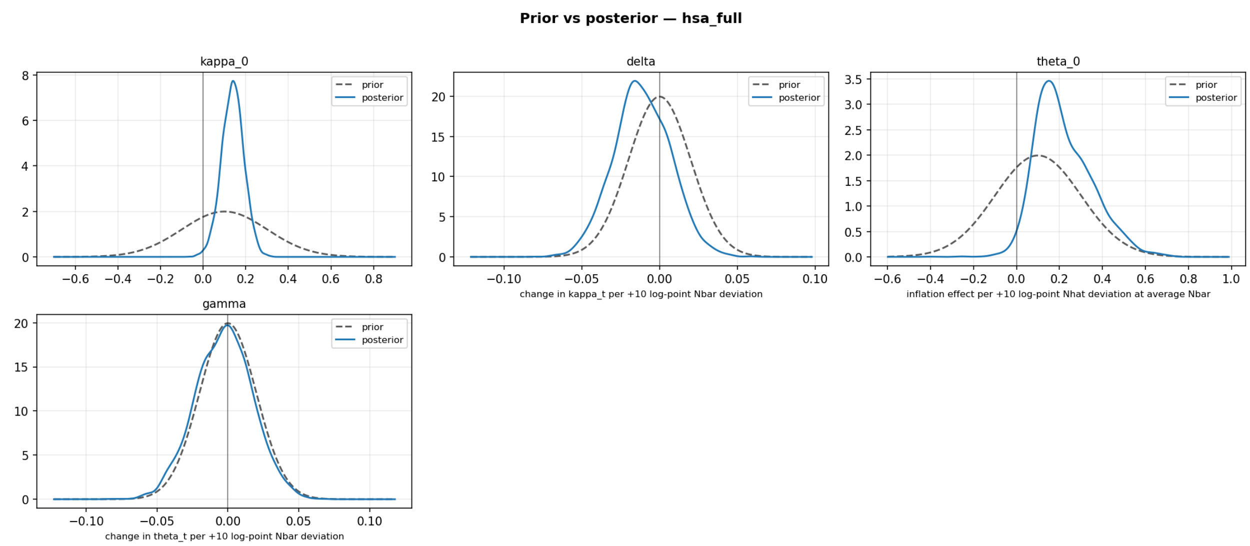
hsa_steady



hsa_dynamic



hsa_full



7.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0282
hsa_dynamic	kappa=0	0.1000	0.2000	0.0459
hsa_dynamic	theta=0	0.1000	0.2000	0.6847
hsa_full	kappa_0=0	0.1000	0.2000	0.1666
hsa_full	delta=0	0.0000	0.0200	0.8608
hsa_full	theta_0=0	0.1000	0.2000	0.2893
hsa_full	gamma=0	0.0000	0.0200	0.9883
hsa_steady	kappa_0=0	0.1000	0.2000	0.1246
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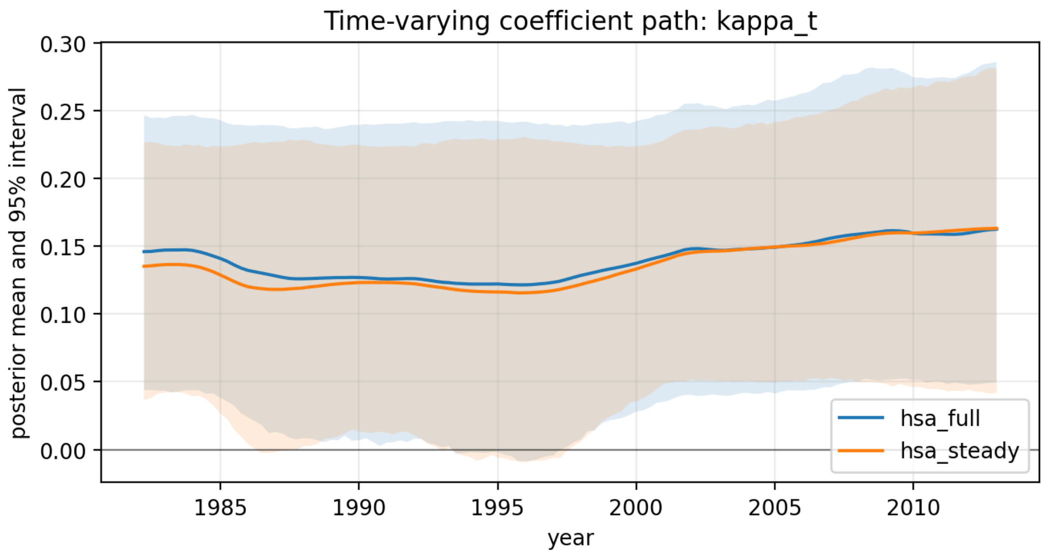

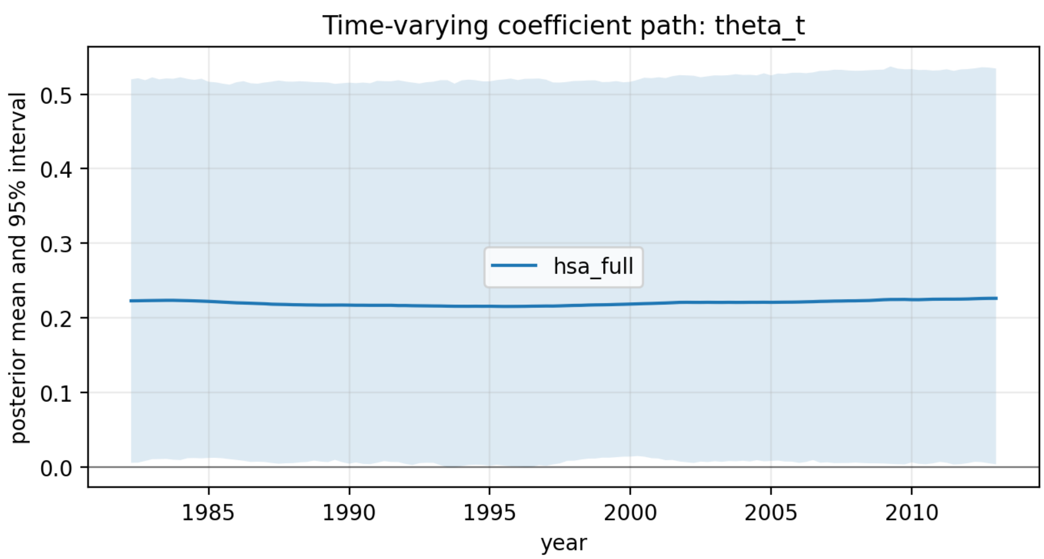

7.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-114.0810	0.6058	-239.3093	1.0000	NaN	NaN	NaN
hsa_dynamic	-112.3597	0.5987	NaN	NaN	NaN	0.6847	NaN
hsa_full	-105.9814	0.5657	NaN	NaN	0.8608	NaN	0.9883
hsa_steady	-113.6351	0.6035	-302.1031	0.0000	1.0043	NaN	NaN

8 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.